25900539 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCNBC200 - NUTRITIONAL BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Gluconeogenesis is the process of: [K] / [CO1]

   a). Breaking down glucose to pyruvate

   b). Formation of glucose from non-carbohydrate sources

   c). Conversion of glycogen to glucose

   d). Oxidation of glucose to CO_2 and water

2. Transmethylation refers to transfer of: [K] / [CO2]

   a). Amino group

   b). Carboxyl group

   c). Methyl group

   d). Phosphate group

3. Triacylglycerolsare composed of…… [K] / [CO3]

   a). Three glycerol molecules and one fatty acid

   b). One glycerol molecule and three fatty acids

   c). One fatty acid and three glycerol molecules

   d). Two glycerol molecules and two fatty acids

4. Under aerobic conditions, pyruvate is converted to [U] / [CO1]

a). Lactate b). Acetyl-CoA

c). Oxaloacetate d). Malate

5. Ketone bodies are primarily synthesized in [U] / [CO3]

a). Brain b). Kidney

c). Liver d). Skeletal muscle

6. The alternate name of HMP shunt is: [U] / [CO1]

a). TCA cycle b). Phosphogluconate pathway

c). Glyoxylate cycle d). Cori cycle

[1x6 = 6]

Part B

Short Answer

Answer any **2** questions

Each question carries **2** marks

7. Explain the classification of deamination reactions [U] / [CO2]

8. Describe the role of thromboxanes in blood clotting and their clinical significance [U] / [CO3]

9. Explain the metabolism of Fructose [U] / [CO1]

10. Define Lactic Acid Cycle [K] / [CO1]

[2x2 = 4]

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

11. Define Glycolysis. [K] / [CO1]

12. Explain any three important reactions in amino acid metabolism with examples. [U] / [CO2]

13. Define β-oxidation of unsaturated fatty acids and list the additional enzymes required. [K] / [CO3]

14. Describe Tricarboxylic Acid. [K] / [CO1]

15. Describe how adipose tissue metabolism contributes to obesity. [U] / [CO3]

16. Describe the energetics of glycolysis with reactions where ATP is consumed and produced. [U] / [CO1]

[5x4 = 20]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

17. Describe ornithine cycle [K] / [CO2]

18. Discuss the role of cholesterol catabolism in the formation of bile acids and bile salts. [K] / [CO3]

19. Describe Glycogenesis and Glycogenolysis [K] / [CO1]

20. Explain the betaoxidation of saturated fatty acid [U] / [CO3]

[10x2 = 20]